RELATIVE PERFECTNESS IS PRESERVED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Relative perfectness is checked on affine opens by the affine-local criterion. invoked at proof lines 9032-9033. Reduces the first proof to affines.
- At the module level relative perfectness is preserved by base change. invoked at proof lines 9034-9035. Concludes the first proof.
- Derived pullback preserves pseudo-coherence. invoked at proof lines 9038-9039. First half of the second proof.
- Tor independence controls the tor dimension of the derived pullback. invoked at proof lines 9040-9041. Second half of the second proof: bounded tor dimension.

Source reference. Original lemma 0DI2: sources/stacks-perfect/pinned-original-source.tex, source lines 8951–8951. Cited at proof lines 9032-9033 to reduce the first proof to affines; the module-level application and conclusion are inside the counted span.

Source reference. Original lemma 0DHW: sources/stacks-chow/tags.txt, source lines 16921–16921. Cited at proof lines 9034-9035 to conclude the first proof; the reduction is inside the counted span.

Source reference. Original lemma 09U7: sources/stacks-chow/tags.txt, source lines 12415–12415. Cited at proof lines 9038-9039 for the pseudo-coherence half of the second proof; the tor-dimension half and conclusion are inside the counted span.

Source reference. Original lemma 0C0V: sources/stacks-perfect/pinned-original-source.tex, source lines 5513–5513. Cited at proof lines 9040-9041 for the bounded tor-dimension half of the second proof; the conclusion is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is flat and locally of finite presentation. Let $g : S' \to S$ be a morphism of schemes. Set $X' = S' \times_S X$ and denote $g' : X' \to X$ the projection. If $K \in D(\mathcal{O}_X)$ is S-perfect, then $L(g')^*K$ is S'-perfect.*

Proof. First proof: reduce to the affine case using Lemma 0DI2 and then apply More on Algebra, Lemma 0DHW.

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Second proof: $L(g')^*K$ is pseudo-coherent by Cohomology, Lemma 09U7 and the bounded tor dimension property follows from Lemma 0C0V. $\square$